12.10 Tying Loose Ends

WORKSHEET

The question

STEP III 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself After tying two ends together, how many pieces of string are left (counting a ring as no longer a "piece")? Does it depend on whether a ring was made?

STAGE 1 The first knot

Covers: the first step

Ask yourself There are $2n$ ends. Pick one: how many of the other ends would make a ring with it?

STAGE 2 All the knots

Covers: the expected number of rings

Pause and try After any knot, ring or not, how many pieces of string are left, each with two ends? So what is the chance of a ring at the knot made when k pieces remain? Add up the expectations.

STAGE 3 The variance

Covers: the variance

Ask yourself Does the chance of a ring at one knot depend on what happened at earlier knots? If not, what can you say about the variances?

STAGE 4 Forty thousand strings

Covers: the case $n = 40\,000$

? **Ask yourself** Write $1 + \frac{1}{3} + \cdots + \frac{1}{2n-1}$ as all reciprocals up to $2n$ minus the even ones. Use $1 + \frac{1}{2} + \cdots + \frac{1}{m} \approx \ln m$.

What's really going on

Logarithmic growth

The expected number of rings grows like $\frac{1}{2}\ln n$, and the variance is close to the mean (both are sums of small probabilities). By the central limit theorem (Section 12.24), the number of rings is approximately normal for large n, with mean and variance about $\frac{1}{2}\ln(4n)$. With 40 000 strings, you will usually get between 2 and 11 rings.

Cycles of random permutations

The same logarithm appears in the number of cycles of a random permutation of n objects: on average $H_n \approx \ln n$ (the strings problem is a close relative, with $\frac{1}{2}\ln n$). Random permutations have few cycles: one giant cycle usually contains most of the elements, and the fraction of elements in the largest cycle averages about 0.62, the Golomb–Dickman constant. Tangling strings at random behaves the same way: most of the string ends up in one or two long loops. These facts are part of the study of *random combinatorial structures*, which has connections to number theory (the prime factors of a random integer behave like the cycles of a random permutation).

Random knots and polymers

The string-tying problem is a toy model for physical systems. Long polymer molecules in a solution join end to end to form rings and chains, and the statistics of loop formation follow similar rules. In knot theory, mathematicians ask the harder question of how likely a random closed loop is to be knotted; for long random loops, the answer tends to 1, a result proved by De Witt Sumners and Stuart Whittington in 1988.

Try this next

1. **Small cases.** For $n = 2$, list the possibilities and check that the expected number of rings is $\frac{4}{3}$.

2. **One big loop.** Show that the probability that the process ends with a single loop containing all the strings is $\prod_{k=2}^{n}\left(1 - \frac{1}{2k-1}\right)$, and that this is approximately $\sqrt{\frac{\pi}{4n}}$ for large n. (*Hint:* relate the product to $\binom{2n}{n}$.)

3. **Cycles of permutations.** Show that the expected number of cycles of a random permutation of n objects is H_n, by building the permutation one element at a time (the "Chinese restaurant" construction).

4. **The variance.** Show that $\mathrm{Var}(R) = \mathrm{E}(R) - \sum_{k=1}^{n} \frac{1}{(2k-1)^2}$, and that the subtracted sum tends to $\frac{\pi^2}{8}$.

Further reading

- **Peter Winkler, *Mathematical Puzzles: A Connoisseur's Collection*** (A K Peters), which includes the string-tying puzzle.
- **Richard Arratia, Andrew Barbour & Simon Tavaré, *Logarithmic Combinatorial Structures*** (European Mathematical Society). Cycles, primes and random structures, for later.
- **Colin Adams, *The Knot Book*** (American Mathematical Society). An introduction to knot theory, including random knots.

The full guided solution is in Section 12.10:
https://beyondstep.keithmcnulty.org/ch12/loose-ends.html